

Phase 3 converts every file in the archive into enriched text before embedding. Scanned pages pass through PP-OCRv6 detection and recognition, while digital PDFs keep their native text layer intact. The Gemma 4 model then answers one structured question battery per document: does the image contain text, is it a chart, who is present, and where are the logical chunk boundaries for retrieval?

Each chunk of 256 to 1024 tokens receives its own vector from the embeddinggemma-300m sidecar running on CPU port 8081. The document summary gets one additional vector for hierarchical search. Hybrid retrieval fuses BM25 over the enrichment JSON with dense cosine similarity, using reciprocal rank fusion at query time.

Throughput is bounded by the enrichment call, not the OCR stage. The server ceiling near 200 tokens per second implies roughly two to six documents per minute during bulk backfill of the family archive.